AERO50006 Mathematics 2, Tutorial Sheet 1 (Signals&Systems part)

1. Use Euler's relation to prove each of the following:

   i $\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$

   ii $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$

$e^{i\theta} = \cos\theta + i\sin\theta$

(i) $e^{i\theta} - e^{-i\theta} = [\cos\theta + i\sin\theta] - [\cos(-\theta) + i\sin(-\theta)]$

$= (\cos\theta + i\sin\theta) - (\cos\theta - i\sin\theta) = 2i\sin\theta$

$\therefore e^{j\theta} - e^{-j\theta} = 2j\sin\theta$

$\therefore \sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$

(ii) $e^{i\theta} + e^{-i\theta} = 2\cos\theta$

$\therefore (e^{i\theta} + e^{-i\theta})^2 = 4\cos^2\theta$

& $(e^{i\theta} + e^{-i\theta})^2$

$= (e^{i\theta})^2 + 2e^{i\theta}e^{-i\theta} + (e^{-i\theta})^2$

$= [\cos^2\theta + 2i\cos\theta\sin\theta - \sin^2\theta] + 2[\cos^2\theta + \sin^2\theta] + [\cos^2\theta - 2i\cos\theta\sin\theta - \sin^2\theta]$

$= 1 + 2\cos^2\theta - 2\sin^2\theta = 2 + 2\cos^2\theta - (1 - \cos 2\theta)$

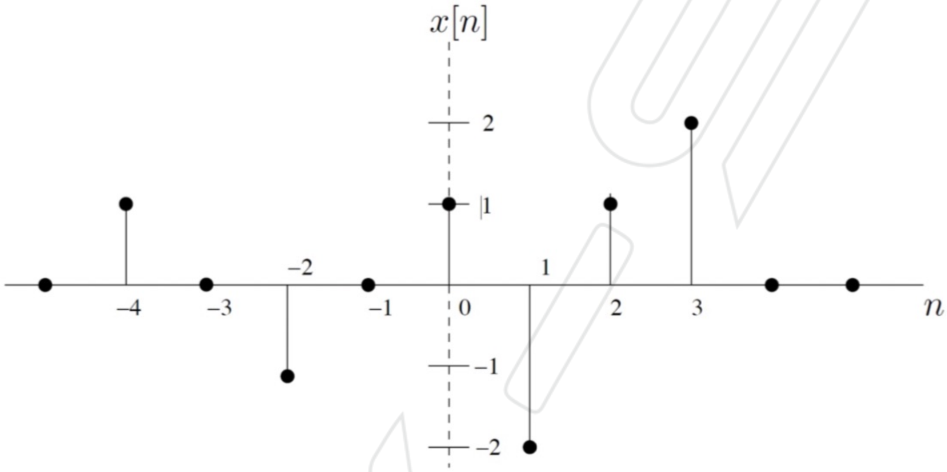
$\therefore \cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$

2. Find and sketch the even and odd parts of each of the following signals. Also find the energy of each (for a CT signal the energy is defined as $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$ and for a DT signal as $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$).

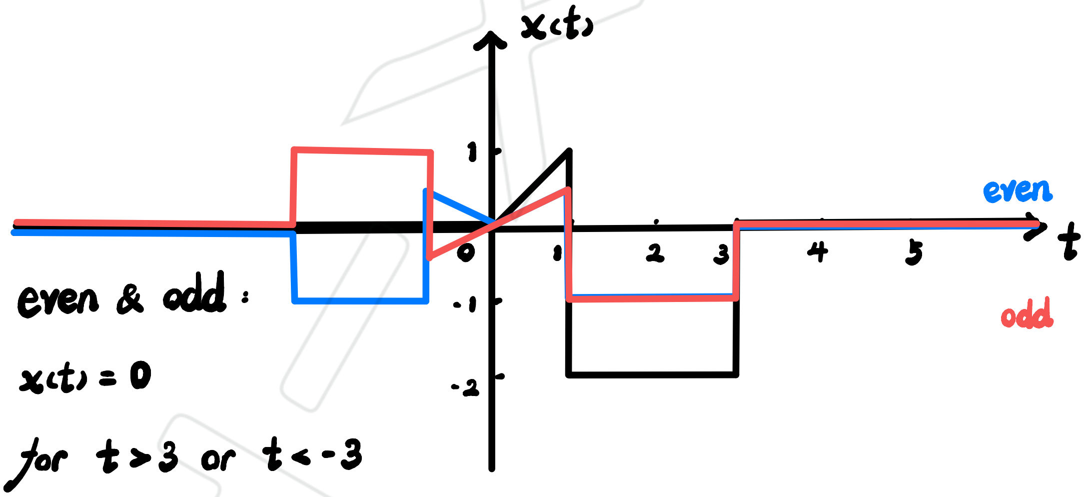
(i.)

$$x(t) = \begin{cases} t & 0 < t \leq 1 \\ -2 & 1 < t \leq 3 \\ 0 & otherwise \end{cases}$$

(ii.) $x[n]$ as shown in the figure below:



(i)



even & odd :

$x(t) = 0$

for $t > 3$ or $t < -3$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^1 t^2 dt + \int_1^3 |-2|^2 dt$$

$$= \left[\frac{1}{3}t^3\right]_0^1 + \left[4t\right]_1^3$$

$$= \frac{1}{3} + (12 - 4) = \frac{25}{3}$$

(ii) odd : $n = \pm 2$

even & odd : $n > 4$ or $n < -4$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$= |1|^2 + |0|^2 + |-1|^2 + |0|^2 + |1|^2 + |-2|^2 + |1|^2 + |2|^2$$

$$= 8 \quad 12$$

3. In the lectures, we defined a linear system as one that is both *scalable* (homogeneous) and *additive*. Determine whether each of the systems below is scalable and/or additive. Justify your answers either with a proof or a counterexample.

(i.)

$$y(t) = \begin{cases} \frac{1}{x(t)} \left[\frac{dx(t)}{dt}\right]^2 & if\ x(t) \neq 0 \\ 0 & otherwise \end{cases}$$

(ii.) $y[n] = \mathcal{R}\{x[n]\}$

(i)

$$x_1 \rightarrow y_1 = \begin{cases} \frac{1}{x_1} \left(\frac{d(x_1)}{dt}\right)^2 & \text{if } x_1 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$x_2 \rightarrow y_2 = \begin{cases} \frac{1}{x_2} \left(\frac{d(x_2)}{dt}\right)^2 & \text{if } x_1 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore\ ay_1 \rightarrow \begin{cases} a\left[\frac{1}{x_2} \left(\frac{d(x_1)}{dt}\right)^2\right] & \text{if } x_1 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{ax_1} \left(\frac{d(ax_1)}{dt}\right)^2 = a\left[\frac{1}{x_1} \left(\frac{d(x_1)}{dt}\right)^2\right] & \text{if } x_1 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$\therefore$ scalable

$\therefore x_3 = ax_1 + bx_2 \longrightarrow$

$$y_3 = \begin{cases} \frac{1}{x_3}\left(\frac{d(x_3)}{dt}\right)^2 = \frac{1}{ax_1+bx_2}\left[\frac{d(ax_1+bx_2)}{dt}\right]^2 & \text{if } x_3 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\neq ay_1 + by_2 \begin{cases} \frac{a}{x_1}\left(\frac{d(x_1)}{dt}\right)^2 + \frac{b}{x_2}\left(\frac{d(x_2)}{dt}\right)^2 & \text{if } x_1, x_2 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$\therefore$ not additive $\quad \therefore$ non-linear

(ii) $x_1 \longrightarrow y_1 = R\{x_1\}$

e.g. $x(t) = 1+2j$

$\downarrow$

not scalable

$x_2 \longrightarrow y_2 = R\{x_2\}$

$ax_1 \longrightarrow ay_1 = aR\{x_1\} = R\{ax_1\}$ $\quad \therefore$ scalable

$x_3 = ax_1 + bx_2 \longrightarrow$

$y_3 = R\{x_3\} = R\{ax_1 + bx_2\} = R\{ax_1\} + R\{bx_2\}$

$= ay_1 + by_2 = R\{ax_1\} + R\{bx_2\}$ $\quad \therefore$ additive

$x_1(t) + x_2(t) \xrightarrow{H} R\{x_1(t) + x_2(t)\}$

$= R\{x_1(t)\} + R\{x_2(t)\}$

$\therefore$ linear

4. In the lectures, we defined a number of general system properties. In particular, we said that a system can be characterized as one or more of the following:

- A Memoryless
- B Causal
- C Time-invariant
- D Linear
- E Stable

For each of the following systems, determine which of the above properties hold. Justify your answers. For each, assume that $u(t)$ is the input and $y(t)$ is the output.

i $y(t) = \int_{-\infty}^{2t} u(\tau)d\tau$,

ii $y(t) = \sin(x(t))$.

(i) A ×

B ✓ ×

C ✓ ×

$$u_1(t) = u(t-\delta)$$

$$y_1(t) = \int_{-\infty}^{2t} u(\tau-\delta)\,d\tau \;\overset{\xi = \tau-\delta}{=\!=\!=}\; \int_{-\infty}^{2t} u(\xi)\,d\tau$$

$$= \int_{-\infty}^{\overset{2t-\delta}{2(t-\delta)}} u(\xi)\,d\xi \neq y(t-\delta)$$

D ✓

$$ay_1(t) = a\int_{-\infty}^{2t} u_1(\tau)\,d\tau = \int_{-\infty}^{2t} au_1(\tau)\,d\tau \quad \therefore \text{scalable}$$

$$u_3(t) = au_1(t) + bu_2(t) \longrightarrow$$

$$y_3(t) = \int_{-\infty}^{2t} u_3(\tau)\,d\tau = \int_{-\infty}^{2t} au_1(\tau) + bu_2(\tau)\,d\tau$$

$$= a\int_{-\infty}^{2t} u_1(\tau)\,d\tau + b\int_{-\infty}^{2t} u_2(\tau)\,d\tau = ay_1(t) + by_2(t) \quad \therefore \text{additive}$$

E ×

$|u(t)| < K$

$|y(t)| = \left|\int_{-\infty}^{2t} u(\tau)\, d\tau\right| = \left|\left[\frac{1}{2}u^2(\tau)\right]_{-\infty}^{2t}\right|$

$= \left|\frac{1}{2}u^2(2t) - \frac{1}{2}u^2(-\infty)\right|$

(iii) A ✓

B ✓

C ✓

$x_1(t) = x(t-\delta)$

$y_1(t) = \sin(x(t-\delta)) = y(t-\delta)$

D ×

$ay_1 = a\sin(x(t)) \neq \sin(ax(t))$ $\quad \therefore$ not scalable

$x_3 = ax_1 + bx_2 \longrightarrow$

$y_3 = \sin(x_3(t)) = \sin(ax_1(t) + bx_2(t))$

$\neq ay_1 + by_2 = \sin(ax_1(t)) + \sin(bx_2(t))$ $\quad \therefore$ not additive

E ✓

$|x(t)| \in \mathbb{R}$

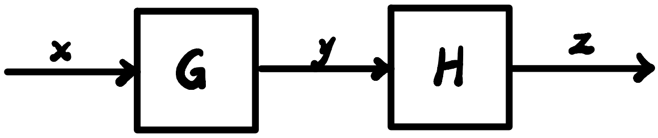
$|y(t)| = |\sin(x(t))|$ $\quad \therefore -1 < |y(t)| < 1$

5. Evaluate the truth of each of these statements. If it is true, provide a proof. If not, provide a counterexample.

   i The series interconnection of two linear systems is always linear.

   ii The series interconnection of two nonlinear systems is always nonlinear.

(i) True



$$G: \sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

$$H: \sum_{k=0}^{0} c_k \frac{d^k z(t)}{dt^k} = \sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k}$$

$$\therefore \sum_{k=0}^{0} c_k \frac{d^k z(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

(ii) False

e.g.

$$G: \sum_{k=-2}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

$$H: \sum_{k=0}^{0} c_k \frac{d^k z(t)}{dt^k} = \sum_{k=-2}^{N} a_k \frac{d^k y(t)}{dt^k}$$

$$\therefore \sum_{k=0}^{0} c_k \frac{d^k z(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

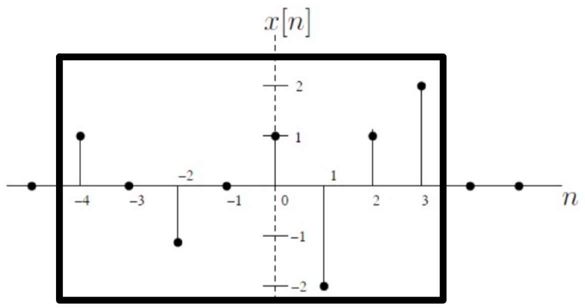
6. For each of the following, calculate and sketch:

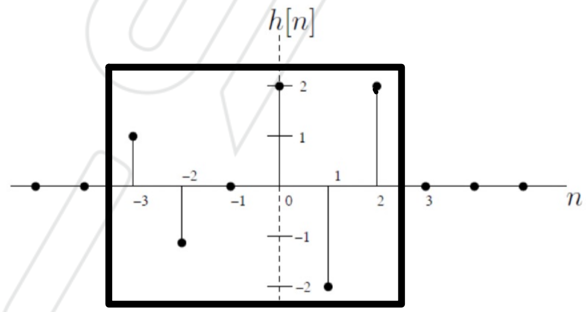
$$y[n] = x[n] * h[n].$$

(i.)

$$x[n] = \begin{cases} \left(\frac{1}{3}\right)^n & n \geq 0 \\ 0 & otherwise \end{cases} \qquad h[n] = \begin{cases} -1 & 0 \leq n \leq 10 \\ 0 & otherwise \end{cases}$$

(ii) $x[n]$ and $h[n]$ as shown below:



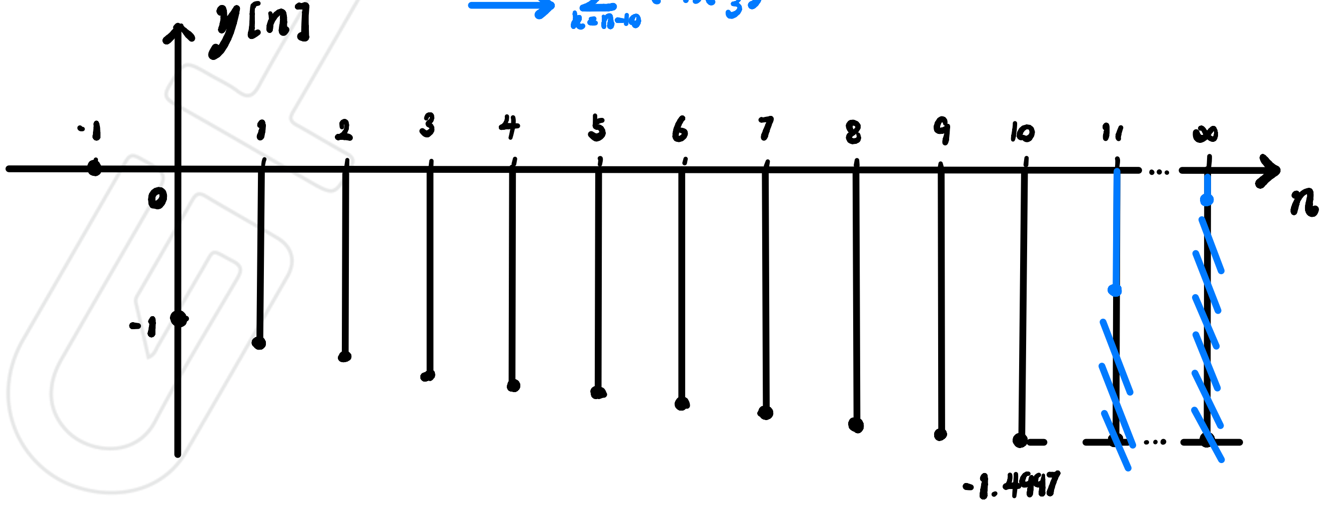


(i) $y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$

$\therefore y[n] = \sum_{k=0}^{n} x[k]\, h[n-k]$ for $0 \leq n \leq 10$

$$\therefore y[n] = \begin{cases} 0 & n < 0 \\ \sum_{k=0}^{n} -\left(\frac{1}{3}\right)^k = -\dfrac{\left(\frac{1}{3}\right)^{n+1} - 1}{\frac{1}{3} - 1} & 0 \leq n \leq 10 \\ \dfrac{-\left(\frac{1}{3}\right)^{n+1} - 1}{\frac{1}{3} - 1} + \dfrac{\left(\frac{1}{3}\right)^{n-9} - 1}{\frac{1}{3} - 1} & n > 10 \end{cases}$$

$\longrightarrow \sum_{k=n-10}^{n} (-1)\left(\frac{1}{3}\right)^k$

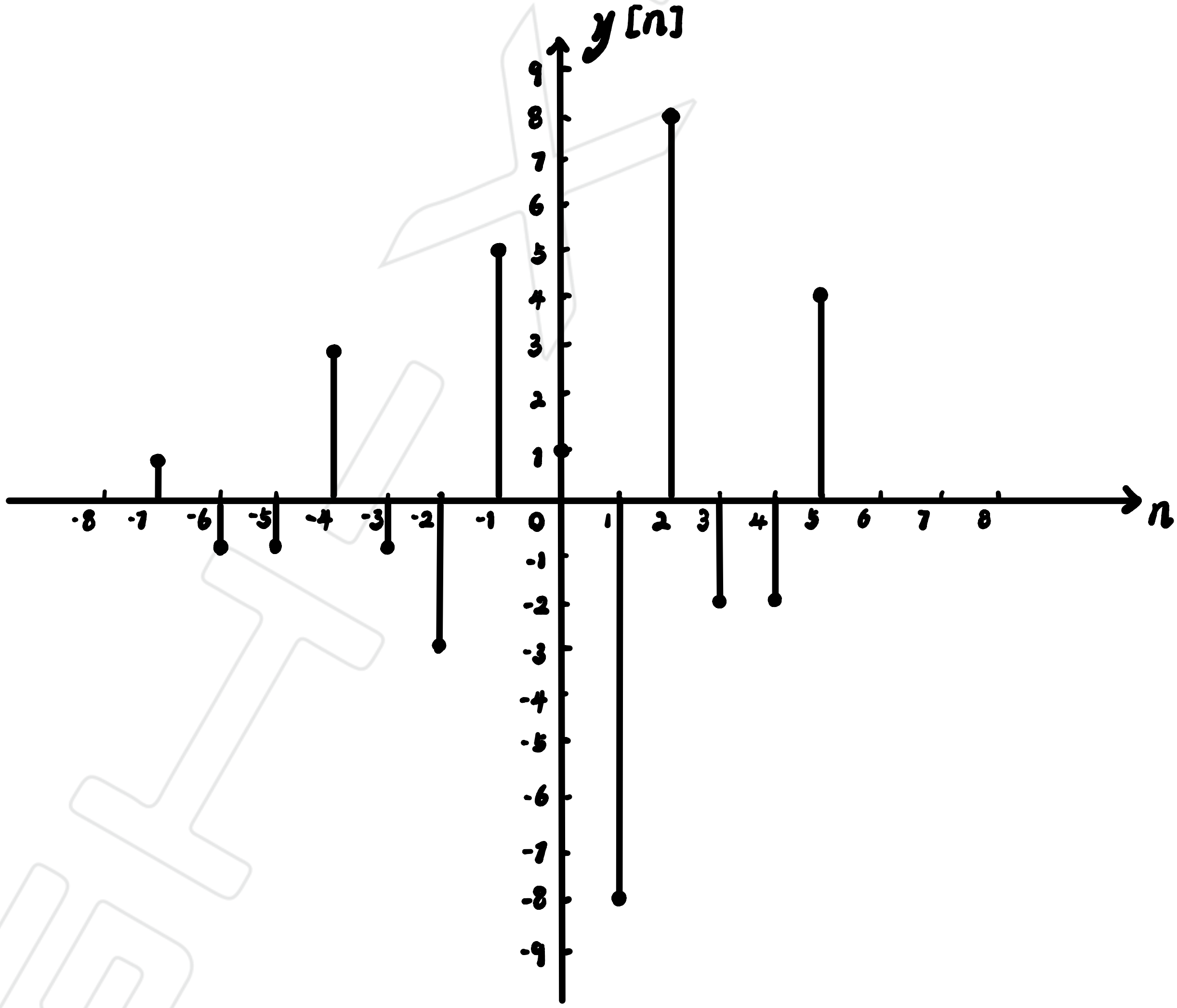


(ii) $y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$

$$\therefore y[n] = \begin{cases} 0 & n < -7 \\ \sum_{k=-4}^{n} x[k]h[n-k] & -7 \leq n \leq 5 \\ 0 & n > 5 \end{cases}$$

$\therefore$ from -7 to 5.

$y[n] = 1, -1, -1, 3, -1, -3, 5, 1, -8, 8, -2, -2, 4$



7. For each of the following, calculate and sketch:

$$y(t) = x(t) * h(t)$$

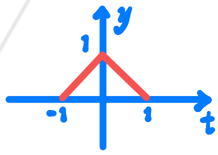
(a)

$$x(t) = \begin{cases} -1 & -1 \le t \le 1 \\ 0 & otherwise \end{cases} \qquad h(t) = \begin{cases} t & 0 \le t \le 1 \\ (2-t) & 1 \le t \le 2 \\ 0 & otherwise \end{cases}$$

(b)

$$x(t) = \begin{cases} \sin(\pi t) & 0 \le t \le 2 \\ 0 & otherwise \end{cases} \qquad h(t) = \begin{cases} 1 & 1 \le t \le 2 \\ 0 & otherwise \end{cases}$$

(c)



$$\begin{cases} 2 & t-2k=0 \\ 0 & t-2k \ne 0 \end{cases}$$

可忽略转置

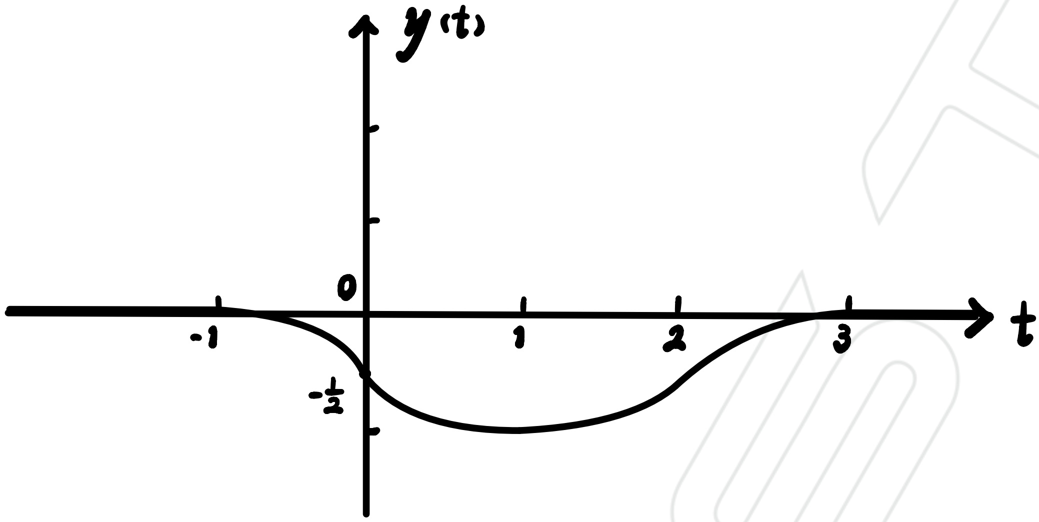
$$x(t) = 2\sum_{k=-\infty}^{\infty} \boxed{\delta}(\underline{t-2\kappa}) \qquad h(t) = \begin{cases} (1+t) & -1 \le t \le 0 \\ (1-t) & 0 \le t \le 1 \\ 0 & otherwise \end{cases}$$

对称

(a) $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

$$\therefore y(t) = \begin{cases} 0 & t \le -1 \\ \int_{-1}^{t} \tau - t \, d\tau & -1 < t \le 0 \\ \int_{t-1}^{t} \tau - t \, d\tau + \int_{-1}^{t-1} -\tau + t - 2 \, d\tau & 0 < t \le 1 \\ \int_{t-1}^{1} \tau - t \, d\tau + \int_{t-2}^{t-1} -\tau + t - 2 \, d\tau & 1 \le t < 2 \\ \int_{t-2}^{1} -\tau + t - 2 \, d\tau & 2 < t \le 3 \\ 0 & t > 3 \end{cases}$$

$$\therefore y(t) = \begin{cases} 0 & t \le -1 \\ -\frac{1}{2}t^2 - t - \frac{1}{2} & -1 < t \le 0 \\ \frac{1}{2}t^2 - t - \frac{1}{2} & 0 < t \le 2 \\ -\frac{1}{2}t^2 + 3t - \frac{9}{2} & 2 < t \le 3 \\ 0 & t > 3 \end{cases}$$



(b) $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

$$\therefore y(t) = \begin{cases} 0 & t \le 1 \\ \int_0^{t-1} \sin(\pi\tau) d\tau & 1 < t \le 2 \\ \int_{t-2}^{t-1} \sin(\pi\tau) d\tau & 2 < t \le 3 \\ \int_{t-2}^{2} \sin(\pi\tau) d\tau & 3 < t \le 4 \\ 0 & t > 4 \end{cases}$$

$$\therefore y(t) = \begin{cases} 0 & t \le 1 \\ -\frac{1}{\pi}[\cos(\pi(t-1)) - 1] & 1 < t \le 2 \\ -\frac{1}{\pi}[\cos(\pi(t-1)) - \cos(\pi(t-2))] = \frac{2}{\pi} & 2 < t \le 3 \\ -\frac{1}{\pi}[1 - \cos(\pi(t-2))] & 3 \le t < 4 \\ 0 & t > 4 \end{cases}$$

(c) $x(t) = 2\sum_{k=-\infty}^{\infty} \delta(t-2k)$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$\boxed{\begin{aligned} &= \int_{0}^{1} 2\sum_{k=-\infty}^{\infty} \delta(\tau-2k)(1+t-\tau)\, d\tau \\ &+ \int_{-1}^{0} 2\sum_{k=-\infty}^{\infty} \delta(\tau-2k)(1-t+\tau)\, d\tau \end{aligned}}$$

$$\int_{-\infty}^{\infty} h(\tau)\,\delta(\tau-2k)\, d\tau = h(2k)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau = 2\sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(\tau-2k)\, h(t-\tau)\, d\tau$$

$$= 2\sum_{-\infty}^{\infty} h(t-2k)$$

$$= \begin{cases} 2\sum_{k=-\infty}^{\infty} (1+t-2k) & -1 \le t-2k \le 0 \\ 2\sum_{k=-\infty}^{\infty} (1-t+2k) & 0 \le t-2k \le 1 \\ 0 & \text{otherwise} \end{cases}$$

8. Suppose that a continuous-time LTI system has impulse response $h(t)$. If the input to this system is a unit step, show that the output is

$$y(t) = \int_{-\infty}^{t} h(\tau)d\tau.$$

This signal is called the *step response* of the system. **NB:** Like the impulse response, the step response can be used to completely characterize the behaviour of the system with respect to any other input.

$$y(t) = x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)\,d\tau$$

$$x(t) = \begin{cases} 0 & t < 0 \\ \frac{t}{\varepsilon} & 0 \le t < \varepsilon \\ 1 & t \ge \varepsilon \end{cases} \xrightarrow{\varepsilon \to 0} x(t-\tau) = \begin{cases} 1 & \tau \le t \\ 0 & \tau > t \end{cases}$$

$$\boxed{y(t) = \begin{cases} \int_{-\infty}^{t} h(t)\,d\tau & t < 0 \\ \int_{-\infty}^{t} h(t)\,d\tau & 0 \le t \le \varepsilon \\ \int_{-\infty}^{t} h(t)\,d\tau + 0 & t \ge \varepsilon \end{cases} \qquad \therefore y(t) = \int_{-\infty}^{t} h(t)\,d\tau}$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$$

$$= \int_{-\infty}^{\infty} x(t-\tau)h(\tau)\,d\tau$$

$$= \int_{-\infty}^{t} x(t-\tau)h(\tau)\,d\tau + \int_{t}^{\infty} x(t-\tau)h(\tau)\,d\tau$$

$$= \int_{-\infty}^{t} h(\tau)\,d\tau$$

9. Show that if the impulse response of an LTI system satisfies the following condition, then the system is BIBO stable:

$$\int_{-\infty}^{\infty} |h(t)|dt < \infty$$

A system that satisfies the above condition is called *absolutely integrable* (*summable* for DT systems).

If $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

for unit step.

$$y(t) = x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

$$x(t) = \begin{cases} 0 & t < 0 \\ \frac{t}{\varepsilon} & 0 \le t < \varepsilon \\ 1 & t \ge \varepsilon \end{cases}$$

$$y(t) = \begin{cases} \int_{-\infty}^{t} h(t) d\tau & t < 0 \\ \int_{-\infty}^{t} h(t) d\tau & 0 \le t \le \varepsilon \\ \int_{-\infty}^{t} h(t) d\tau + 0 & t \ge \varepsilon \end{cases}$$

$\therefore y(t) = \int_{-\infty}^{t} h(t) d\tau$

$\therefore |y(t)| < \infty$ $\quad \therefore$ BIBO stable

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A system that satisfies the above condition is called *absolutely integrable* (*summable* for DT systems).

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau$$

$$\therefore |y(t)| = \left| \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau \right| \le \int_{-\infty}^{\infty} |x(t-\tau) h(\tau)| d\tau$$

$$\le \int_{-\infty}^{\infty} |x(t-\tau)| |h(\tau)| d\tau$$

let $|x(t-\tau)| \le M$

$$\therefore |y(t)| \le \int_{-\infty}^{\infty} M |h(\tau)| d\tau$$

$$\therefore |y(t)| \le M \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

$$\because \int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

$\therefore$ BIBO stable